

DHRUV MOTAVAL

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI / GenAI: LangGraph, LangChain, RAG, OpenAI API, Groq API, LangSmith

LLMs: GPT, Claude Opus, Claude Sonnet, Qwen, Llama, DeepSeek, GLM, Fable, Ollama, llama.cpp, GGUF

Backend: FastAPI, REST APIs, Pydantic, SQLite, PostgreSQL, ChromaDB, Sentence Transformers

Voice & NLP: Twilio Voice, Whisper, ElevenLabs, spaCy

DevOps & Tools: Docker, Git, GitHub Actions (CI/CD), Railway, Vercel

EXPERIENCE

Web Developer Intern - iWebwala

Jan 2026 – Mar 2026

Python, FastAPI, Twilio, Whisper, ElevenLabs, Railway, Netlify, Git

- Built and deployed a production conversational AI assistant using Twilio Voice, Whisper, ElevenLabs, and FastAPI, automating approximately **50% of inbound customer inquiries**.
- Integrated** telephony, speech recognition, and LLM services through **FastAPI REST APIs**, connecting voice interactions into a unified production workflow.
- Collaborated with client stakeholders to refine requirements and deploy production updates on **Railway**.

PROJECTS

NomadicOS - Local-First AI Operating Environment

github.com/dhruv-motaval/nomadicos

Python, LangGraph, Ollama, llama.cpp, GGUF, Pydantic

- Developed a local-first AI runtime with **Python and LangGraph**, combining LLM reasoning, **structured object-type memory and RAG retrieval**, Action IR validation, authorization, OS execution, and goal verification.
- Implemented controlled OS execution with owner authorization, revocation epochs, single-use authorization, **Windows filesystem confinement**, and structured terminal execution.
- Validated** the runtime with a **203-test regression suite and 7 live local-model tests**, covering execution controls, process cleanup, and evidence-based verification.

RAG Document Q&A System

github.com/dhruv-motaval/rag-document-qa

FastAPI, LangChain, ChromaDB, Sentence Transformers, LangSmith, SQLite

- Developed** an end-to-end **RAG document Q&A system** using **FastAPI, LangChain, ChromaDB, and Sentence Transformers**, covering PDF ingestion, embeddings, vector search, and grounded generation.
- Implemented 2 retrieval-dependent response paths** through similarity-threshold guardrails, returning grounded answers when evidence is sufficient and “Not found in document” otherwise.
- Measured 25 ms average retrieval latency** across **50+ LangSmith-traced evaluation queries** and deployed the system as a live product.

Real-Time Multilingual Call Resolution System

github.com/dhruv-motaval/Real-Time-Multilingual-Call-Resolution-MyOnSite

LangGraph, FastAPI, Whisper, spaCy, Groq, SQLite

- Designed** a **4-stage LangGraph workflow** for transcription, entity extraction, intent classification, and conversation routing across **3 languages: English, Hindi, and Spanish**.
- Automated 2 workflow actions** through LLM tool-calling: ticket creation and follow-up scheduling, exposed through structured **FastAPI REST APIs**.
- Established** deterministic offline validation using mocked **Whisper, spaCy, and LLM clients**, enabling end-to-end testing without live API dependencies.

EDUCATION

B.Tech, Artificial Intelligence - Parul Institute of Technology, Vadodara, Gujarat

2023 – 2026

CGPA: 7.12 / 10

Diploma, Computer Engineering - A. Y. DadaBhai Institute of Technology, Kosamba, Gujarat

2019 – 2023

CGPA: 7.91 / 10